



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

COURSE INFORMATION SHEET
(For Theory Based Course)

Session:	Fall-2024
Course Title:	Web Engineering
Course Code:	CS-311T
Credit Hours:	2
Semester:	5 th
Pre-Requisites:	Programming Fundamentals
Instructor Name:	Mr. Mohsin Naqvi
Email and Contact Information:	mohsin.naqvi@ssuet.edu.pk
WhatsApp Group	Web Engg Fall 2024
Office Hours:	Monday to Friday (08:30am to 04:30pm)
Mode of Teaching:	Synchronous / Asynchronous / Hybrid / Blended

COURSE OBJECTIVE:

The goal of the course is to give knowledge about web engineering concepts, principles process, methods and models. How to model design for web applications and what models' perspective are useful for construction. Familiarize with Web application development software tools and environments currently available on the market. Build Web Applications that are scalable flexible to modify and easy to manage.

COURSE OUTLINE:

Web programming languages, Design principles of Web based applications, Web platform constraints, Web standards, Responsive Web Design, Web Applications, Browser/Server Communication, Storage Tier, Cookies and Sessions, Input Validation, Full stack state management, Web App Security, Large scale applications, Performance of Web Applications, Enabling Technologies for adoption of cloud services and future Data Centers, Web Testing and Web Maintenance.



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COURSE OUTCOMES:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	Explain the basic concepts, processes, and standards related to the web engineering for web application design.	PLO_1 (Academic Education)	C2 (Understanding)
2	Describe models and techniques to develop static and responsive web applications.	PLO_2 (Knowledge for solving Computing Problems)	C2 (Understanding)
3	Apply web standards and techniques to develop reliable and secure web application	PLO_3 (Problem Analysis)	C3 (Applying)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	6 (14%)	4 (14%)	-----
Assignments	6 (14%)	4 (14%)	-----
Mid	20 (48%)	10 (36%)	-----
Final	10 (24%)	10 (36%)	30(100%)
Total	42 (42%)	28 (28%)	30 (30%)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	100%	100

Recommended Book:

- **Web Engineering the Discipline of Systematic Development of Web Applications** by Gerti Kappel, Birgit Proll, Siegfried Reich, Werner R. ISBN 13: 9780470015544
- **Web Engineering**, Rajiv Chopra, Prentice-Hall of India, 2016. Publisher: Prentice-Hall of India Pvt.Ltd; 1st edition (September 30, 2016), ISBN-13: 978-8120352544.

Reference Books:

- **Web Engineering: Modelling and Implementing Web Applications (Human-Computer Interaction Series)** 2014 Edition by Rossi ISBN 978-1-84628-923-1



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- **Web Engineering a Practitioners Approach** – 2010 by PRESSMAN ISBN 9780073523293
- **Current Trends in Web Engineering** ICWE 2018 International Workshops, 2018

LECTURE PLAN

Course Title: Web Engineering

Course Code: CS-311T

Week No.	Week Dates	Topics	Required Reading	Key Dates
1	08-10-2024 to 11-10-2024	Introduction to Web Engineering, World Wide Web, design, development, and evaluation of web applications	Gerti Kappel Chap 01 Pages 01-16	
2	14-10-2024 to 18-10-2024	Client-side programming, Making interactive web pages and concept of Back-end services	Gerti Kappel Chap 02 Pages 23-36	Assignment 1
3	21-10-2024 to 25-10-2024	Building Blocks of web Applications, Front-end programming, HyperText Markup Language (HTML5)	Gerti Kappel Chap 03 Pages 39-43	Quiz 1
4	28-10-2024 to 01-11-2024	Cascading Style Sheets (CSS3), Presentation of a document written in HTML or XML	Gerti Kappel Chap 03 Pages 44-58	
5	04-11-2024 to 08-11-2024	JavaScript, why JavaScript is important, behavior of web pages, examples of JavaScript programs	Gerti Kappel Chap 03 Pages 59-63	Assignment 2
6	11-11-2024 to 15-11-2024	Back-end programming, what does a back-end developer do, back-end programming languages, Hypertext Processor (PHP), Java & JSP, ASP.Net, C++, Python, Ruby	116 https://en.wikipedia.org/wiki/Single-page_application	Quiz 2
7	18-11-2024 to 22-11-2024	Design principles of Web based applications, Navigation, Consistency, Visual hierarchy	Gerti Kappel Chap 06 Pages 111-116	
8	25-11-2024 to 29-11-2024	Web platform constraints, Software as a Service (SaaS), rules and limitation defined in web design and development	Gerti Kappel Chap 06 Pages 126-133	
9	Midterm Examination (02-12-2024 to 07-12-2024)			



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10	09-12-2024 to 13-12-2024	Web standards, responsive web design, web applications and benefits, web services	Gerti Kappel Chap 04 Pages 65-71	
11	16-12-2024 to 20-12-2024	Browser/Server Communication, Storage Tier, Cookies and Sessions	Gerti Kappel Chap 05 Pages 72-89	
12	23-12-2024 to 27-12-2024	Testing web Apps (Methods and Techniques, Test Plan and Test Generation)	Gerti Kappel Chap 07 Pages133-152	
13	30-12-2024 to 03-01-2025	Web Testing (Selenium), Web App Security - Browser Isolation, Network Attacks, Session Attacks	Gerti Kappel Chap 07 Pages133-152	Assignment3
14	08-01-2024 to 12-01-2024	Large scale web applications, deciding the complexities of large scale web application development	Gerti Kappel Chap 11 Pages219-243	Quiz 3
15	06-01-2025 to 10-01-2025	Performance of Web Applications, quality and efficiency of web application functions	Gerti Kappel Chap 12 Pages247-263	
16	13-01-2025 to 17-01-2025	Data Centers, Web Testing and Web Maintenance, types of web application testing	Gerti Kappel Chap 13 Pages265-291	
Final Examination (28-01-2025 to 08-02-2025)				